

4th Downs Decoded: A Selection-Bias-Corrected Study of Player Impact on Fourth Down Conversions in the NFL

Simon Raymond¹ and Nick Manuel

simon.raymond@uwaterloo.ca

nicholas.manuel@smu.ca

¹ Corresponding Author

Abstract

This research breaks down fourth downs in the NFL by identifying crucial player positions and evaluating how coaches incorporate different factors in their decision-making process. Using a newly engineered dataset that includes detailed player statistics² from 2017-2023, we analyze the causal factors influencing fourth down decisions and outcomes. Our non-linear sample selection correction reveals that coaches factor in unobserved variables during fourth down decisions that can predict the likelihood of a fourth down conversion. With selection bias accounted for, we identify the key player characteristics that influence a 4th down conversion. We further observe that coaches are making the decision to attempt a 4th down based on player characteristics that are not crucial to a 4th down.

Keywords: NFL, decision making, fourth down, player analysis, selection bias

JEL Codes: Z2, Z29

² We thank Pro Football Focus (PFF) for providing proprietary player grade data for this research.

Section 1: Introduction

It is the 2023-2024 NFL season; the Detroit Lions have twenty-two minutes and three seconds between them and their first Super Bowl appearance. They lead the San Francisco 49ers 24 to 10 with the ball on their opponents 28-yard line. However, it is fourth down and the Lions would need to gain 2 more yards to earn a new set of downs to continue their hopes of scoring yet another touchdown that would add 6 points to their total and surely put the game away. Should the Lions fail this fourth down attempt they turn the ball over to their opponents, in favourable field position. Historically teams would not attempt this 4th down but would rather kick a field goal to gain 3 points or to punt the ball to give the opponent worse field position. Despite this, the Lions and their head coach Dan Campbell have adopted a more aggressive strategy and decide to attempt the 4th down as they have many times that year. Thus the Lions line up in shotgun formation; the ball is snapped to the Quarterback who avoids pressure and throws an incomplete pass. Despite leading in score, this failed fourth down attempt seemed to send the Lions into a downward spiral and they ended up losing the game 31 to 34.

While Dan Campbell's aggressive fourth down attempt was viewed as reckless by many fans, it represents a leaguewide shift of coaches moving away from their cautious ways by listening to analytics. Romer (2006) and Yam and Lopez (2018) both suggest that NFL teams

need to be more aggressive on fourth downs. However, these traditional analytics cannot incorporate that NFL coaches know their players, their motivations and the momentum of the game while we do not. On top of that there is selection bias as fourth downs that are not attempted are never visible to us. This has prevented researchers from properly looking at what variables can impact the outcome of a fourth down.

Thus, in this paper we first begin by confirming if there is a selection bias present on fourth downs. For this we implement a methodology similar to Das et al (2003) in the form of a two-stage selection-bias correction that has a non-parametric first stage and a polynomial correction term. We are in the fortunate situation in which we can verify the validity of our exclusion restriction which is the “time remaining in the half” through the bias-free channel of third downs. We find significance of our correction term on the conversion of a fourth down which implies that coaches incorporate variables into their decision making that are not visible to researchers.

With bias being corrected for on fourth downs we now ask what types of NFL teams succeed on fourth downs more often. To answer this question, we construct a data set that has statistics and player grades of all relevant players from each team merged into play-by-play data. These statistics are highly specific, such as a Quarterback’s grades when throwing the ball short/medium/long distances which allows us to identify what player types help their teams the most on 4th down. This granularity lets us find that players that excel in run blocking are significantly more valuable when compared to players that specialize more in the passing game.

Finally, we seek to answer if coaches let the quality of their players influence their attempt decisions. To this we find evidence that coaches do let player quality influence their behaviour. However, they often let the wrong types of player attributes influence their decision making. For example, our results find that defensive rush grades heavily affect the conversion probability. However, coaches do not pick up on that variable in their decision to attempt rather they listen more to the pass rush grades of the defenders.

With the NFL reaching new heights of competitiveness, it places even greater importance on fourth downs as they create huge swings in expected points. Our research allows coaches to build their teams and align their decisions with the driving factors of fourth down conversion. Most importantly however, we have provided empirical proof that there are moments in which an NFL coach should trust their instincts as they observe variables that no researcher sitting behind a computer can view.

Our paper proceeds as follows. Section 2 provides a literature review in which we discuss publications on coaches' decision making, player evaluation, and selection bias correction methodologies. Section 3 describes how we merge weekly player grades into play-by-play data and presents some basic summary information about it. Section 4 outlines a two-stage model that corrects for selection bias and justifies our restrictive variable. Empirical proof of exogeneity and strength for this exclusion restriction is given in section 5 which present that and other results that pertain to our corrective terms and player grades. Lastly, we make some concluding remarks and attach an appendix with the information on how our non-parametric first stage was tuned.

Section 2: Literature Review

Much discourse revolves around the idea that NFL coaches are acting overly averse to risk, which is lowering their expected wins. Romer (2006) found that teams had begun to move towards a more conservative strategy in the NFL. He argues that teams value successful gambles more than the expected win percentage in a game. This is to say that a coach favours punting and kicking more than attempting a fourth down as they are easier to be seen as “successful”. He theorizes that the poor decision making is either due to risk aversion or poor information. To further this point using matching analysis Yam and Lopez (2018) quantified this conservative decision-making, finding that teams could gain approximately 0.4 wins per year by being more aggressive on fourth downs. Goff and Locke (2019) found when revisiting Romer’s framework that Romer’s core findings are still held to be true. However, they argue that overly conservative calls are not due to poor decision making. Instead, they point to risk aversion as they estimate that coaches are willing to give up two-thirds of an expected point to avoid the uncertainty of fourth down attempts. On top of this, there seems to be evidence that coaches are more cautious when their job is on the line. Owens and Roach (2018) found that in the NCAA coaches are relatively more conservative when they are more likely to be fired. At the same time, they found when a coach was likely to be promoted, they behaved more aggressive than normal.

The evaluation of players in the NFL has seen a push away from measuring traditional measures such as yards or receptions as not every yard has the same value (Carter and Machol, 1971). Work from Burke (2009) popularized the use of Expected Points Added. Yurko et al.

(2019) proposed a nflWAR framework that introduced a novel multinomial logistic regression approach for estimating expected points. They were also able to identify players that excelled in different situations (short vs. long passing). However, they are limited to QBs, WRs, TEs, and RBs in their analysis.

We apply a version of Heckman's (1979) sample selection correction to correct for any biases created from missing 4th down data points that are never attempted. The model consists of the selection and outcome stage. The first step is to estimate the probability of observations being selected into the outcome stage sample. A probit model that assumes normal errors is used in this first stage. The probabilities from this model are then converted using the Inverse Mills Ratio which again assumes normality. Das et al (2003) showed that a non-parametric selection step model can be employed to account for non-linear relationships in the selection stage. They then further examine the different ways of constructing the selection correction term that accounts for non-normality with polynomials and splines instead of a classic Inverse Mills Ratio.

Previous papers that have done some form of analysis in NFL data use team aggregate statistics to measure the quality of a team. While some researchers, such as Yurko et al. (2019), have used player statistics in their analysis, they were limited to skill position statistics such as touchdowns, yards gained, or receptions. This completely overlooks the offensive line and defensive positions. We contribute to the literature using a new type of dataset that has information on every player that has the potential to impact a play's outcome. This dataset contains Pro Football Focus (2024) player grades. Finally, we have provided an empirical proof

that the “time remaining in the half” is exogenous to fourth down outcomes and thus may be used as an exclusion restriction when controlling for selection bias.

Section 3: Data

Our data was pulled from two main sources. As a base the nflverse (2023) package in R provided us play-by-play data for the years of 2017 to 2023. This includes basic game information and the IDs of players that participated in each play. The reason for our cutoff of 2017 is due to the availability of data from our second source with PFF. This data, which contains highly detailed information on individual players, is downloaded as premium player reports on a weekly basis and then aggregated to time lengths of 12 weeks. Players from the PFF data set were first merged into our base play-by-play data set based on pre-matched player identification columns provided by the nflverse package. If a player did not have a pre-matched ID then they were matched based on names, teams, and positions. We assign every player to their respective positions and order players who have the same position on the same team by their depth chart rankings which are available via the nflverse package. For example, the starting Quarterback is QB1 while the backup is QB2. While it depends on position, we keep a maximum of three players of a single position to reduce noise as only starters and possibly direct backups would have a chance to influence a game consistently.

Our measures of player performance are Pro Football Focus player grades. Pro Football Focus assigns individual players specific grades for their performance during each play. These grades are based on their expert observations, which are especially useful for positions that are not easily measured with quantifiable statistics like offensive linemen. They also look to distinguish a player’s individual performance from the performance of their teammates. Table 1

shows what specific grades we use for each position and how many players at each position are included in our data.

Table 1: Breakdown of all 116 Performance Grades Tracked by Position

Pos	Position Name	Players	Grades Recorded	Total Per Player
QB	Quarterback	1	Pass Short, Pass Medium, Pass Deep	3
RB	Running Back	1-3	Pass Block, Rush Run, Receive Short Route, Receive Medium Route, Receive Deep Route	5
WR	Wide Receiver	1-3	Run Block, Receive Short Route, Receive Medium Route, Receive Deep Route	4
TE	Tight End	1-3	Pass Block, Run Block, Receive Short Route, Receive Medium Route, Receive Deep Route	5
T	Tackle	1-3	Pass Block, Run Block	2
G	Guard	1-3	Pass Block, Run Block	2
C	Center	1-2	Pass Block, Run Block	2
DE	Defensive End	1-3	Run Defense, Pass Rush	2
DT	Defensive Tackle	1-3	Run Defense, Pass Rush	2
NT	Nose Tackle	1	Run Defense, Pass Rush	2
ILB	Inside Linebacker	1-3	Run Defense, Man Coverage, Zone Coverage	3
OLB	Outside Linebacker	1-3	Run Defense, Man Coverage, Zone Coverage	3
CB	Cornerback	1-3	Run Defense, Man Coverage, Zone Coverage	3
FS	Free Safety	1-2	Run Defense, Man Coverage, Zone Coverage	3
SS	Strong Safety	1-2	Run Defense, Man Coverage, Zone Coverage	3
K	Kicker	1	Field Goals/Extra Points	1
P	Punter	1	Punting	1

Notes: Players column shows the range of player numbers tracked for each position (e.g., 1-3 means positions 1, 2, and 3) All grades represent 12-week rolling averages

Our measure of a player’s abilities at the outset of a game is a rolling average of their PFF performance statistics during the 12 previous weeks. A 12-week window is selected to reduce noise that may arise from short windows while simultaneously avoiding use of outdated performance measures. For robustness, we test whether our results hold using two weeks window. We also find that a 12-week window preforms better on out of bag test sets. These results are reported in the robustness checks subsection of our paper.

Alongside these variables we include a “Missing Information” variable for each position/grade. These are binary indicators that are = 1 if a player’s grades were not available due to the player making their first start or not having a recorded grade of that specific variable.

It should also be noted that we scrape the number of fans in attendance from each game from Pro Football Reference (2024). Below are some basic summary statistics of our data when prepared for use within our models.

Table 2: Descriptive Statistics of Key Game Variables

Variable	4th Down Attempt			4th Down Conversion			3rd Down Conversion		
	Mean	Median	SD	Mean	Median	SD	Mean	Median	SD
4th Down Attempt	0.16	0.00	0.37	—	—	—	—	—	—
4th Down Converted	—	—	—	0.55	1.00	0.50	—	—	—
3rd Down Converted	—	—	—	—	—	—	0.40	0.00	0.49
Rush Attempt	—	—	—	0.40	0.00	0.49	0.22	0.00	0.41
Week	9.84	10.00	4.84	10.19	10.00	4.88	9.79	10.00	4.84
Yards to Go	7.81	7.00	5.71	3.46	2.00	3.90	6.99	6.00	5.01
Yardline	50.54	55.00	24.66	35.27	37.00	20.72	50.24	54.00	24.22
Off. Timeouts	2.69	3.00	0.58	2.53	3.00	0.72	2.72	3.00	0.56
Def. Timeouts	2.69	3.00	0.61	2.61	3.00	0.65	2.71	3.00	0.58
Temperature	61.58	70.00	15.23	60.68	70.00	15.34	61.71	70.00	15.02
Wind	5.39	4.00	5.74	5.35	4.00	6.00	5.29	4.00	5.72
Vegas Win Prob.	0.47	0.45	0.32	0.35	0.25	0.33	0.50	0.50	0.31
Spread Line	-0.35	-1.00	6.57	-0.90	-2.00	6.61	-0.02	-1.00	6.55
Total Line	44.89	44.50	4.62	45.31	45.00	4.58	45.09	45.00	4.62
Prep Days	7.48	7.00	2.11	7.47	7.00	2.13	7.47	7.00	2.09
Score Differential	-1.05	0.00	10.34	-6.45	-7.00	11.82	-1.08	0.00	10.31
Home Attendance	59704.69	67175.00	23226.52	59536.33	67322.00	23859.93	59334.14	66975.00	23586.78
Overall Win Pct.	0.53	0.54	0.21	0.52	0.50	0.21	0.54	0.56	0.21
Team Win Pct.	0.43	0.42	0.13	0.42	0.42	0.13	0.43	0.42	0.13
Kicker Grade	61.74	62.05	4.02	59.97	61.93	11.02	61.77	62.10	4.06
Punter Grade	61.35	61.27	2.31	60.59	61.10	6.44	61.35	61.27	2.30
Home Team	0.49	0.00	0.50	0.48	0.00	0.50	0.50	0.00	0.50
First Half	0.48	0.00	0.50	0.34	0.00	0.48	0.49	0.00	0.50
Seconds in Half	915.46	900.00	476.74	762.16	700.00	465.35	929.12	916.00	484.47
Formation: Shotgun	—	—	—	0.51	1.00	0.50	0.71	1.00	0.45
Formation: Singleback	—	—	—	0.19	0.00	0.39	0.06	0.00	0.24
Formation: I-Form	—	—	—	0.08	0.00	0.27	0.03	0.00	0.17
Formation: Empty	—	—	—	0.11	0.00	0.31	0.14	0.00	0.35
Formation: Wildcat	—	—	—	0.01	0.00	0.09	0.00	0.00	0.06
Formation: Jumbo	—	—	—	0.05	0.00	0.22	0.01	0.00	0.10
Formation: Pistol	—	—	—	0.02	0.00	0.13	0.01	0.00	0.11
Defense Personnel: DL	—	—	—	3.20	3.00	1.16	2.90	3.00	1.13
Defense Personnel: LB	—	—	—	2.74	3.00	1.09	2.71	3.00	1.10
Defense Personnel: DB	—	—	—	4.71	5.00	1.21	5.15	5.00	1.04
Defense Personnel: Other	—	—	—	0.03	0.00	0.17	0.02	0.00	0.15

Data Dimensions (after filtering): 4th Down Attempt: 20236 obs, 387 vars; 4th Down Conversion: 3137 obs, 413 vars; 3rd Down Conversion: 36765 obs, 413 vars. Note: Table excludes 116 missing player information indicator variables, 116 individual player performance variables, and 80 coach indicator variables for brevity.

Section 4: Methodology

$$E[\text{Conversion}_i | x_i, \text{Attempt}_i = 1] = h_0(c_i) + g_0(x_i) + \lambda_0(\hat{p}_i) \quad (1)$$

$$\hat{p}_i = P(\text{Attempt}_i = 1 | c_i, x_i, z_i) = \text{XGBoost}(c_i, x_i, z_i) \quad (2)$$

Equation 1 contains our model for the probability of a successful 4th down conversion on play i . c_i contains our measures of player performance listed in Table 1. x_i contains a rich array of relevant control variables listed in Table 2 that are predictive of successful 4th down conversion attempts. Despite our extensive set of controls, the non-random nature of 4th down conversions attempts creates a standard sample selection problem in the style of Heckman (1979). We only observe the success or failure of 4th down attempts if a 4th down conversion was actually attempted. Thus coaches may be more or less likely to attempt a 4th down conversion based on unobservable variables that effect the likelihood of success. In this case our estimated effects of player characteristics on 4th down conversions will be biased if the unobservable information is correlated with our player characteristics.

To properly account for sample selection bias, we non-parametrically estimate $\hat{p}_i$ as the probability of a conversion attempt on play i . Our non-parametric model of choice to estimate $\hat{p}_i$ is XGBoost developed by Chen & Guestrin (2016). XGBoost (Extreme Gradient Boosting) is a machine learning algorithm that builds an ensemble of decision trees (or parametric regressions) to capture non-linear relationships to make predictions. See appendix A for tuning procedure and results.

Our exclusion restriction in $\mathbf{z}_i$ contains a quadratic function of the number of seconds remaining in the half, as well as its interaction with the score differential between the offensive and defensive teams on play i , and an indicator for whether the offensive team is winning. The exogeneity of seconds remaining to the conversion of 4th down attempts is apparent as in each of these 4th down attempts the goal of the attempt is to simply gain a new set of downs or, if the

team is in the redzone, to score. The time left till the game or half ends cannot directly influence the physical execution of the play itself, but may affect the coach's willingness to accept risk.

We use a non-parametric sample selection equation shown in Equation 2 to obtain our sample correction term $\lambda_0(\hat{p}_i)$, where $\hat{p}_i$ is the estimated probability of a 4th down attempt as obtained after estimating Equation 2. Our exclusion restriction z_i includes a quadratic function for seconds remaining in the half, alongside its interactions with the score differential and an indicator for whether the offensive team is winning at the time of the play. We exclude plays i that occur within the final two minutes of each half since during these late time periods coaches are increasingly likely to attempt to score on play i rather than simply look for a first down to continue the drive. Coaches' risk tolerance change as they approach the end of the half. For most of the game, their immediate objective is to convert to a new set of downs. In the last two minutes of a game teams look score quickly. These fundamental difference in coaches' decisions during the dying seconds of the game makes these plays unrepresentative of those that occur for the majority of the game. The exact trimming point of 2 minutes was chosen as it is a natural cutoff due to the "2-minute warning" that is imposed each half of a game.

To test the exogeneity of z_i , we examine whether it is able to successfully predict conversions on 3rd down. When teams reach 3rd down, they always attempt to convert to a new set of downs, meaning the sample selection bias is not a problem in this context. While 3rd down attempts are not identical to 4th down attempts, as the coach must consider the future possibility of special team usage, they possess the same dire "last chance" trait as a 4th down. This presents a rare opportunity where we may test the exogeneity of our restrictive variables in z_i in a highly

similar sample that does not have any selection bias problem, meaning that if z_i is unable to predict 3rd down conversion success, it is unlikely to predict 4th down conversion success. The first three columns of Table 3 report the estimated marginal effects of the listed independent variable on the probability of a 3rd down conversion success, using linear probability model, probit and logit estimations. As a robustness check, we also report the corresponding values when the final three minutes and final five minutes of each half are trimmed. We consistently find no relationship between seconds remaining in half and the probability of a successful 3rd down conversion.

On the other hand, the last three columns of Table 3 show that the effect of z_i on the decision to attempt a 4th down is highly significant with p-values well under 0.01. This effect comes from the different times in which a coach is willing to accept risk throughout a game.

Once we obtain $\hat{p}_i$ from Equation 2, we proceed as recommended by Das et al (2003) by using the inverse mills ratio in the gaussian case as our estimated propensity to attempt a 4th down in Equation 1. This will capture the normality that is within the true functional form of λ_0 . We then include polynomials to account for non-linearities.

$$\lambda_i = \varphi(\Phi^{-1}(\hat{p}_i))/\hat{p}_i \quad (3)$$

Where: $\varphi(\cdot)$ = standard normal PDF and $\Phi(\cdot)$ = standard normal CDF

Then by polynomial expansion we define

$$\lambda_0(\hat{p}_i) = \gamma_1 \lambda_i + \gamma_2 \lambda_i^2 + \gamma_3 \lambda_i^3 \quad (4)$$

Where γ_i represents the coefficients on λ_i^j

Section 5: Results

Table 3: Marginal Effects of Seconds Remaining in the Half on Third and Fourth Downs

Variable	Third Down Conversion			Fourth Down Attempt		
	LPM	Probit	Logit	LPM	Probit	Logit
(>2 minutes remaining)						
<i>T</i>	-0.011 (0.019)	-0.013 (0.021)	-0.012 (0.021)	-0.052*** (0.018)	-0.044*** (0.010)	-0.027*** (0.007)
<i>T</i> ₂	0.003 (0.018)	0.004 (0.020)	0.004 (0.020)	0.056*** (0.016)	0.042*** (0.010)	0.028*** (0.007)
<i>SD</i>	0.004 (0.022)	0.006 (0.024)	0.005 (0.025)	-0.202*** (0.023)	-0.080*** (0.011)	-0.062*** (0.008)
<i>W</i>	-0.019 (0.032)	-0.021 (0.035)	-0.019 (0.035)	-0.014 (0.028)	-0.027** (0.013)	-0.013 (0.010)
<i>W</i> × <i>SD</i>	-0.032* (0.019)	-0.035 (0.021)	-0.036* (0.022)	0.137*** (0.018)	0.047*** (0.010)	0.034*** (0.007)
<i>T</i> × <i>SD</i>	0.006 (0.035)	0.004 (0.040)	0.007 (0.040)	0.214*** (0.039)	0.055*** (0.019)	0.048*** (0.014)
<i>T</i> × <i>W</i>	0.056 (0.036)	0.062 (0.040)	0.060 (0.040)	0.085*** (0.030)	0.066*** (0.018)	0.041*** (0.013)
<i>T</i> ² × <i>SD</i>	-0.003 (0.024)	-0.002 (0.027)	-0.003 (0.028)	-0.057** (0.026)	-0.005 (0.013)	-0.008 (0.009)
<i>T</i> ² × <i>W</i>	-0.034 (0.026)	-0.037 (0.029)	-0.037 (0.029)	-0.068*** (0.021)	-0.048*** (0.013)	-0.030*** (0.009)
<i>T</i> × <i>W</i> × <i>SD</i>	-0.044 (0.038)	-0.049 (0.042)	-0.048 (0.042)	-0.132*** (0.034)	-0.041** (0.019)	-0.032** (0.014)
<i>T</i> ² × <i>W</i> × <i>SD</i>	0.040 (0.028)	0.044 (0.031)	0.044 (0.031)	0.047** (0.024)	0.012 (0.013)	0.010 (0.010)
(>3 minutes remaining)						
<i>T</i>	-0.004 (0.021)	-0.007 (0.023)	-0.005 (0.023)	-0.015 (0.018)	-0.019** (0.009)	-0.010* (0.006)
<i>T</i> ₂	-0.001 (0.020)	0.001 (0.022)	-0.001 (0.022)	0.023 (0.017)	0.021** (0.008)	0.012** (0.006)
<i>SD</i>	-0.002 (0.024)	0.002 (0.027)	-0.000 (0.027)	-0.229*** (0.026)	-0.085*** (0.011)	-0.063*** (0.008)
<i>W</i>	0.028 (0.038)	0.028 (0.042)	0.034 (0.042)	0.029 (0.032)	-0.006 (0.015)	0.003 (0.011)
<i>W</i> × <i>SD</i>	0.045*** (0.022)	-0.050** (0.024)	-0.052** (0.025)	0.133*** (0.020)	0.040*** (0.009)	0.029*** (0.007)
<i>T</i> × <i>SD</i>	0.008 (0.041)	0.004 (0.046)	0.009 (0.047)	0.248*** (0.044)	0.070*** (0.019)	0.057*** (0.013)
<i>T</i> × <i>W</i>	0.010 (0.042)	0.014 (0.046)	0.010 (0.047)	0.041 (0.034)	0.036** (0.018)	0.018 (0.012)
<i>T</i> ² × <i>SD</i>	-0.003 (0.027)	-0.001 (0.031)	-0.004 (0.031)	-0.076*** (0.028)	-0.019 (0.012)	-0.017* (0.009)
<i>T</i> ² × <i>W</i>	-0.006 (0.029)	-0.008 (0.032)	-0.006 (0.033)	-0.040* (0.023)	-0.028** (0.012)	-0.014* (0.008)
<i>T</i> × <i>W</i> × <i>SD</i>	-0.017 (0.044)	-0.018 (0.049)	-0.017 (0.049)	-0.132*** (0.039)	-0.037** (0.018)	-0.028** (0.013)
<i>T</i> ² × <i>W</i> × <i>SD</i>	0.023 (0.031)	0.025 (0.035)	0.025 (0.035)	0.045* (0.026)	0.011 (0.013)	0.008 (0.009)
(>5 minutes remaining)						
<i>T</i>	0.004 (0.024)	0.003 (0.027)	0.005 (0.027)	0.011 (0.021)	-0.002 (0.007)	-0.000 (0.005)
<i>T</i> ₂	-0.007 (0.023)	-0.006 (0.026)	-0.008 (0.026)	-0.000 (0.020)	0.005 (0.007)	0.003 (0.005)
<i>SD</i>	-0.034 (0.030)	-0.036 (0.034)	-0.038 (0.035)	-0.219*** (0.032)	-0.067*** (0.011)	-0.048*** (0.007)
<i>W</i>	0.050 (0.053)	0.058 (0.059)	0.062 (0.061)	0.030 (0.045)	-0.004 (0.015)	0.005 (0.012)
<i>W</i> × <i>SD</i>	-0.032 (0.029)	-0.037 (0.032)	-0.037 (0.033)	0.128*** (0.027)	0.031*** (0.010)	0.020*** (0.007)
<i>T</i> × <i>SD</i>	0.060 (0.055)	0.065 (0.062)	0.069 (0.063)	0.255*** (0.060)	0.063*** (0.018)	0.048*** (0.012)
<i>T</i> × <i>W</i>	-0.005 (0.057)	-0.008 (0.063)	-0.011 (0.063)	0.033 (0.046)	0.023 (0.017)	0.008 (0.012)
<i>T</i> ² × <i>SD</i>	-0.033 (0.035)	-0.036 (0.039)	-0.038 (0.040)	-0.080** (0.036)	-0.020** (0.010)	-0.016** (0.007)
<i>T</i> ² × <i>W</i>	0.002 (0.038)	0.004 (0.042)	0.006 (0.042)	-0.032 (0.030)	-0.018 (0.011)	-0.007 (0.008)
<i>T</i> × <i>W</i> × <i>SD</i>	-0.048 (0.060)	-0.049 (0.067)	-0.050 (0.068)	-0.139** (0.054)	-0.033* (0.018)	-0.019 (0.013)
<i>T</i> ² × <i>W</i> × <i>SD</i>	0.042 (0.040)	0.044 (0.045)	0.045 (0.045)	0.048 (0.034)	0.011 (0.012)	0.005 (0.009)

Note:

Displays marginal effects with HC1 robust SEs in parentheses.

Variables: T = Seconds Remaining, SD = Score Difference, W = Offensive Team Winning.

* p < 0.1, ** p < 0.05, *** p < 0.01

Selection Bias Correction Terms

With our restrictive variable, and therefore correction terms, justified we estimate our primary model. In our outcome equation we find our sample selection correction terms are all significant at a p-value of 0.10. This implies selection bias on fourth downs. Thus, coaches observe and act upon variables that are unobservable to analysts that in turn effect the probability of converting on a fourth down.

Table 4: IMR Correction Terms ($\lambda_0(\hat{p}_i)$)

Variable	LPM	Probit	Logit
IMR Correction Terms			
IMR (Linear)	-0.367 (0.233)	-0.551 (0.293)*	-0.607 (0.315)*
IMR (Polynomial 2nd order)	0.647 (0.398)	1.033 (0.547)*	1.141 (0.598)*
IMR (Polynomial 3rd order)	-0.307 (0.182)*	-0.519 (0.286)*	-0.574 (0.320)*
Model Controls			
Situational Controls	Yes	Yes	Yes
Game & Team Controls	Yes	Yes	Yes
Weather Controls	Yes	Yes	Yes
Coach Fixed Effects	Yes	Yes	Yes
Play-Type Controls	Yes	Yes	Yes

Notes:

Table displays Marginal Effects for IMR terms and indicates inclusion of control groups

* p < 0.1, ** p < 0.05, *** p < 0.01

Player Characteristics

With selection bias accounted for, we estimate the effect of player characteristics on the likelihood of successful fourth down conversions. In addition to our outcome models on the conversion of a 4th down we have included similar models that explain the probability of attempting a fourth down. These will allow us to compare what player qualities matter on a 4th down relative to the characteristics that are influencing coaches' attempt decisions.

Due to there being 116 player variables that are all interacted with if the play is a run play in our conversion model, we have included Table 5 that displays the joint significance p-values for each position. Note that the player characteristics for defensive positions are those for on-field players of the defensive team during the play, rather than characteristics of the offensive teams' off-field defensive players. LPM models use a Wald-Test while the probit and logit models use a Likelihood Ratio Test. After consulting these joint tests for where to look we can then break down the marginal effect of different variables of the player individually.

Table 5: P-Values for Joint Significance Tests - Individual Position Groups

Variable	Attempt Model			Conversion Model		
	LPM	Probit	Logit	LPM	Probit	Logit
Offensive Positions						
Quarterbacks	0.432	0.735	0.682	0.311	0.148	0.170
Running Backs	0.225	0.410	0.416	0.536	0.167	0.156
Wide Receivers	0.290	0.330	0.292	0.039**	0.008***	0.009***
Tight Ends	0.606	0.738	0.762	0.462	0.161	0.183
Tackles	0.362	0.740	0.563	0.635	0.427	0.413
Guards	0.125	0.220	0.182	0.331	0.059*	0.078*
Centers	0.904	0.524	0.597	0.593	0.420	0.423
Defensive Positions						
Defensive Ends	0.042**	0.090*	0.109	0.656	0.555	0.583
Defensive Tackles	0.029**	0.008***	0.010**	0.126	0.078*	0.068*
Nose Tackles	0.548	0.901	0.957	0.335	0.175	0.188
Inside Linebackers	0.578	0.238	0.332	0.913	0.743	0.826
Outside Linebackers	0.928	0.860	0.758	0.923	0.701	0.719
Cornerbacks	0.597	0.726	0.702	0.161	0.160	0.156
Free Safeties	0.393	0.353	0.249	0.031**	0.020**	0.017**
Strong Safeties	0.007***	0.032**	0.061*	0.183	0.060*	0.077*
IMR Correction Terms						
IMR (Linear)	—	—	—	0.116	0.060*	0.054*
IMR (Polynomial 2nd order)	—	—	—	0.104	0.059*	0.056*
IMR (Polynomial 3rd order)	—	—	—	0.092*	0.070*	0.073*
Model Controls						
Situational Controls	Yes	Yes	Yes	Yes	Yes	Yes
Game & Team Controls	Yes	Yes	Yes	Yes	Yes	Yes
Weather Controls	Yes	Yes	Yes	Yes	Yes	Yes
Coach Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Play-Type Controls (Formation & Run/Pass)	No	No	No	Yes	Yes	Yes

Note:

Each position group tested jointly (e.g., all QBs together, all RBs together).

* p < 0.10, ** p < 0.05, *** p < 0.01

What stands out the most at first is that Quarterbacks' strength is not showing up to be predictive of either 4th down attempts or 4th down conversion success despite being the highest paid position (ESPN , 2025). However, in Table 6, we report a significant differences between

the effects of a QB's short and deep (as well as medium) passing grades. The opposing signs of the effects between short and deep passing grades for both interacted and confounded variables suggest a difference in the usefulness of quarterback archetypes. This is to say that a quarterback that excels with short passes is more useful on 4th down pass attempts due to most pass plays being short passes. However, when the play call is a run, it is more beneficial to have a QB that excels in deep passing as it forces the defense to adjust to the potential threat of a deep pass which takes defensive pressure away from the run game.

Table 6: Marginal Effects of Interacted and Confounded QB Passing Grades on Conversion Probability

Variable	Uninteracted Effects (w/ Rush Attempt)			Interaction Effects (w/ Rush Attempt)		
	LPM	Probit	Logit	LPM	Probit	Logit
QB Passing Grade (Marginal Effects)						
QB Short Pass Grade	0.015 (0.016)	0.023 (0.018)	0.022 (0.019)	-0.022 (0.021)	-0.039 (0.027)	-0.038 (0.028)
QB Medium Pass Grade	-0.028* (0.016)	-0.036* (0.019)	-0.038* (0.019)	0.040* (0.022)	0.057** (0.027)	0.062** (0.028)
QB Deep Pass Grade	-0.019 (0.016)	-0.028 (0.018)	-0.027 (0.019)	0.023 (0.021)	0.035 (0.025)	0.032 (0.026)
Z Test of ME Equality (p-values)						
Short vs Deep Pass Grade	0.124	0.049**	0.070*	0.119	0.043**	0.070*
Short vs Medium Pass Grade	0.054*	0.024**	0.028**	0.039**	0.011**	0.012**
Medium vs Deep Pass Grade	0.694	0.756	0.683	0.580	0.547	0.427

Note:

Table displays marginal effects with standard errors in parentheses. The bottom panel shows p-values from Wald tests comparing the equality of marginal effects.

* * p < 0.10, ** p < 0.05, *** p < 0.01

Another way to approach this is in Table 7 which looks at the joint significances of the different categories of player grades across all relevant positions. This lets us make three broad identifications.

- (1) Blocking grades are found to be statistically significant. In this it is also important that the other more commonly-reported grades such as receiving, rushing or passing are not found to affect the conversion of fourth downs.
- (2) With our conversion model estimates we see defensive rushing grades are found to be significant. This aligns with the significance of offensive run blocking grades.

(3) Coaches fail to attempt a 4th down on the player grades/attributes that matter to the conversion of a fourth down. This comes from the only significant variables at a p-value > 0.10 being the pass rush grades and the zone coverage grades. These are not significant on 4th down conversions.

(4)

We theorize that this mismatch in how coaches make decisions, based on player quality, comes from a misunderstanding of what truly matters on 4th downs. For example, many top defensive linemen are considered elite due to their pass rush ability. While this greatly matters for the full game, it is not an area that is as important on a 4th down when compared to a players run-stopping ability.

Table 7: P-Values for Joint Significance Tests - Grade Types Across Positions

Variable	Attempt Model			Conversion Model		
	LPM	Probit	Logit	LPM	Probit	Logit
Offensive Grade Types						
Pass Blocking (RB, TE, OL)	0.378	0.812	0.667	0.245	0.030**	0.031**
Run Blocking (WR, TE, OL)	0.444	0.503	0.489	0.059*	0.003***	0.004***
Rushing (RB)	0.110	0.153	0.172	0.288	0.137	0.128
Passing (QB)	0.432	0.735	0.682	0.311	0.148	0.170
Receiving/Pass Routes (RB, WR, TE)	0.329	0.430	0.445	0.527	0.172	0.177
Defensive Grade Types						
Run Defense (All Defense)	0.521	0.380	0.558	0.021**	0.003***	0.004***
Pass Rush (DL)	0.002***	0.001***	0.001***	0.679	0.546	0.529
Man Coverage (LB, DB)	0.501	0.695	0.617	0.303	0.130	0.150
Zone Coverage (LB, DB)	0.076*	0.058*	0.079*	0.949	0.950	0.959
IMR Correction Terms						
IMR (Linear)	–	–	–	0.116	0.060*	0.054*
IMR (Polynomial 2nd order)	–	–	–	0.104	0.059*	0.056*
IMR (Polynomial 3rd order)	–	–	–	0.092*	0.070*	0.073*
Model Controls						
Situational Controls	Yes	Yes	Yes	Yes	Yes	Yes
Game & Team Controls	Yes	Yes	Yes	Yes	Yes	Yes
Weather Controls	Yes	Yes	Yes	Yes	Yes	Yes
Coach Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Play-Type Controls (Formation & Run/Pass)	No	No	No	Yes	Yes	Yes

Note:

Attempt models test base grade variables; Outcome models test base variables plus interaction terms with rush attempt.

OL = T+G+C, DL = DE+DT+NT, LB = ILB+OLB, DB = CB+FS+SS.

* p < 0.10, ** p < 0.05, *** p < 0.01

We can further expand by looking at specific positions and the marginal effects of their player grades. The next set of tables shows player grades that are interacted with if the play was

a rush attempt when estimating conversion probability on 4th downs. While we cannot display every single table for brevity, the most important tables have been picked out for deeper analysis and will provide the following link³ for verification that the patterns we discuss hold true for all similar positions.

Table 8 shows that the nose tackle, similar to other defensive positions, demonstrates that players that excel in run defense make it harder for an offense to convert a fourth down. Likewise, a player that excels in pass defense is less effective in stopping the run.

Table 8: Fourth Down Conversion Model Marginal Effects for Nose Tackles

Metric	LPM		Probit		Logit	
	Pass	Rush	Pass	Rush	Pass	Rush
NT 1: Pass Rush	-0.061 (0.114)	0.266 (0.134)**	-0.090 (0.140)	0.370 (0.169)**	-0.095 (0.146)	0.386 (0.177)**
NT 1: Run Defense	0.066 (0.100)	-0.245 (0.133)*	0.069 (0.123)	-0.342 (0.166)**	0.079 (0.128)	-0.354 (0.174)**

Notes: Displays marginal effects with robust SEs in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.10.

Table 9 shows that on the offense we find a similar type of result as the top two wide receivers possess statistically significant run blocking grades.

Table 9: Fourth Down Conversion Model Marginal Effects for Wide Receivers

Metric	LPM		Probit		Logit	
	Pass	Rush	Pass	Rush	Pass	Rush
WR 1: Run Block	-0.013 (0.018)	0.055 (0.024)**	-0.017 (0.020)	0.076 (0.028)***	-0.015 (0.021)	0.078 (0.029)***
WR 1: Deep Route	0.019 (0.019)	0.002 (0.021)	0.025 (0.022)	0.001 (0.026)	0.028 (0.023)	0.002 (0.028)
WR 1: Medium Route	0.032 (0.018)*	-0.042 (0.023)*	0.044 (0.021)**	-0.055 (0.028)*	0.046 (0.022)**	-0.058 (0.030)*
WR 1: Short Route	0.008 (0.017)	0.003 (0.022)	0.010 (0.019)	0.004 (0.026)	0.010 (0.020)	0.009 (0.028)
WR 2: Run Block	-0.030 (0.017)*	0.036 (0.021)*	-0.035 (0.019)*	0.048 (0.024)**	-0.037 (0.019)*	0.050 (0.025)**
WR 2: Deep Route	0.017 (0.020)	-0.010 (0.020)	0.022 (0.023)	-0.014 (0.023)	0.019 (0.024)	-0.012 (0.024)
WR 2: Medium Route	0.003 (0.018)	-0.041 (0.021)*	0.004 (0.021)	-0.059 (0.025)**	0.001 (0.022)	-0.059 (0.027)**
WR 2: Short Route	-0.003 (0.017)	-0.017 (0.019)	-0.004 (0.019)	-0.025 (0.023)	-0.003 (0.020)	-0.029 (0.024)
WR 3: Run Block	-0.007 (0.018)	0.023 (0.020)	-0.000 (0.021)	0.020 (0.026)	-0.001 (0.022)	0.021 (0.027)
WR 3: Deep Route	-0.023 (0.027)	-0.004 (0.021)	-0.025 (0.031)	-0.007 (0.026)	-0.029 (0.033)	-0.010 (0.028)
WR 3: Medium Route	-0.042 (0.028)	0.005 (0.022)	-0.056 (0.033)*	0.003 (0.027)	-0.053 (0.034)	0.003 (0.029)
WR 3: Short Route	-0.018 (0.027)	0.030 (0.024)	-0.030 (0.032)	0.047 (0.028)*	-0.031 (0.033)	0.049 (0.030)*

Notes: Table displays marginal effects with standard errors in parentheses. Standard errors are heteroskedasticity-robust. *** p < 0.01, ** p < 0.05, * p < 0.10.

³ <https://simonraymond2003.github.io/SupplementaryTables/index.pdf>

Thus, wide receivers that perform better when blocking for a run play increase the likelihood of converting on a fourth down rushing attempt. Similarly to quarterbacks, wide receivers with higher medium pass route grades have a negative effect on a team when a rushing play is called. The contrasts between passing and rushing grades point to how certain types of wide receivers are more effective than others on fourth downs.

Two-week Window Robustness Checks

When deciding between a 12 week or 2 week rolling window to evaluate our players on, we ran 10,000 tests on bootstrapped samples of the second stages of each model. In these we confirm that the 2-week window is relatively noisier and thus has a large decrease in predictive performance as seen by difference in mean area under the ROC curve across all simulations.

Figure 1

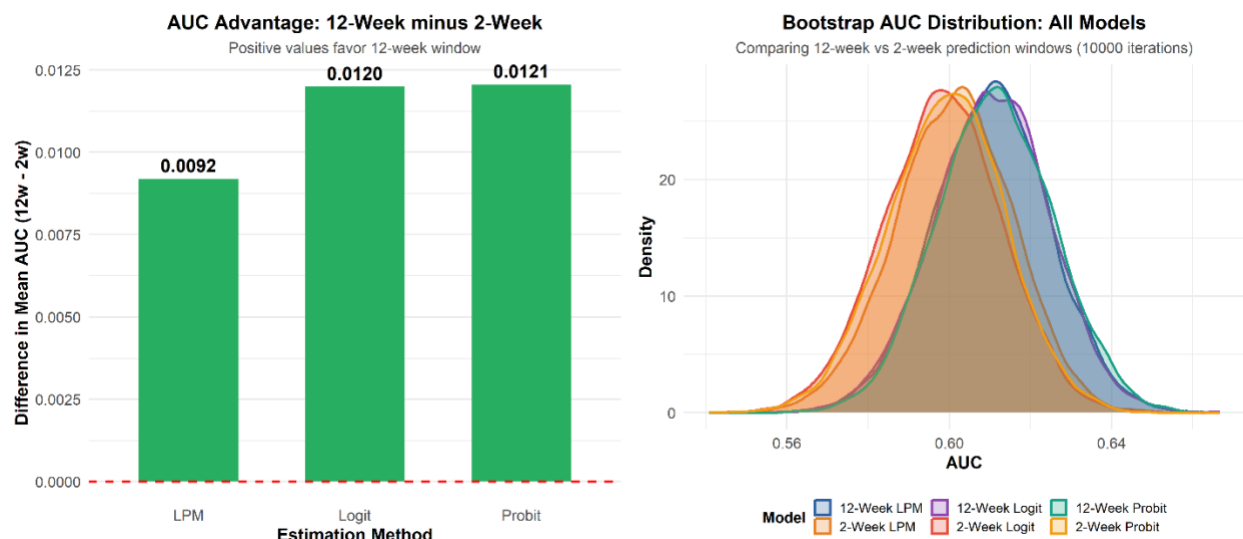


Figure 1

We also can use this 2-week player model to check the robustness of our selection bias correction. Table 10 shows our IMR polynomials maintain their direction and significance.

Table 10: IMR Correction Terms ($\lambda_0(\hat{p}_i)$) of our 2-week and No-Players models

Variable	With Player Variables (2-week)			Without Player Variables		
	LPM	Probit	Logit	LPM	Probit	Logit
IMR Correction Terms						
IMR Linear	-0.406 (0.230)*	-0.580 (0.296)*	-0.623 (0.313)**	-0.456 (0.203)**	-0.558 (0.237)**	-0.562 (0.248)**
IMR Polynomial 2	0.669 (0.394)*	0.977 (0.552)*	1.048 (0.586)*	0.776 (0.338)**	0.963 (0.412)**	0.961 (0.435)**
IMR Polynomial 3	-0.284 (0.182)	-0.437 (0.284)	-0.465 (0.304)	-0.334 (0.150)**	-0.423 (0.196)**	-0.420 (0.210)**
Model Controls						
Situational Controls	Yes	Yes	Yes	Yes	Yes	Yes
Game & Team Controls	Yes	Yes	Yes	Yes	Yes	Yes
Weather Controls	Yes	Yes	Yes	Yes	Yes	Yes
Coach Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Play-Type Controls	Yes	Yes	Yes	Yes	Yes	Yes
Player Variables	Yes	Yes	Yes	No	No	No

Notes: Table displays marginal effects with standard errors in parentheses for IMR terms. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10 also shows that our IMR polynomials are robust to not having specific player variables in the model.

Section 6: Conclusion

The literature on 4th down conversions in football has compared the average value of a conversion attempt relative to punting or attempting a field goal (Romer 2006, Goff and Locke 2018). However, there has been little research to date into what specific team compositions are predictive of successful 4th down conversion attempts. We address this gap in the literature in this paper by using a sample selection correction method to identify the effects of specific positions and player skills that increase the likelihood of a successful 4th down conversion attempt.

We find strong evidence that teams with offensive players who have good pass and run blocking grades are more likely to successfully convert on 4th down attempts, conditional on a wide array of other team and game characteristics. However, we simultaneously find that teams with better pass and rush blocking offenses do not attempt 4th down conversions more often than teams with weaker pass and rush blocking offenses despite this being a notable predictor of the likelihood of success.

We also find that when defensive teams have better run defense scores, the likelihood of a successful 4th down conversion attempt by the offense decreases. Despite this, 4th down attempt decisions appear to be made on the basis of the defenses' pass defense rather than their run defense.

Regarding specific positions, we find that the overall strength of wide receivers and free safeties effect the likelihood of successful 4th down conversion attempts. However, the strength of defensive tackles and strong safeties are most predictive of conversion attempts.

Despite our findings suggesting that NFL coach's 4th down conversion attempt decisions are often made based on player skills that appear to have little relevance to 4th down conversion success, we also find strong evidence that coach's are evaluating and correctly acting upon information that is unobservable to us as researchers. 4th down conversion attempts during game situations when such attempts are less common are more successful at converting, likely in response to information that we can not see but which suggests to the coach that the attempt is relatively likely to be successful.

Reproducibility

All of our codes in R are made available via the following GitHub page⁴. Prior to running the code, you will need to purchase and individually download the PFF premium player grades for each week from 2017 to 2023. Out of respect for PFF's terms of use of the data, data must be obtained directly from them.

Appendix A

To tune our first stage XGBoost model we fix the sets the following hyperparameters as constant and tune the rounds to them.

$$\eta(\text{shrinkage}) = 0.01, \max_depth = 4, \min_child_weight = 1, \text{subsample} = 1, \\ \text{colsample_bytree} = 1, \gamma = 0, \lambda = 1, \alpha = 0$$

For validation splits we use out-of-sample bootstrapping. Our tuning process on the rounds hyperparameter searches from 1 to 2000 in increments of 50. We run each hyperparameter set on a maximum of 150 on validation steps. However, to save computation time, the bottom 1% (rounded up) of hyperparameter sets are dropped after each iteration until we are left with 10 hyperparameters sets. Figure 2 shows that we tune rounds to 851.

Figure 2

⁴ https://github.com/SimonRaymond2003/4th_downs_reproducibility

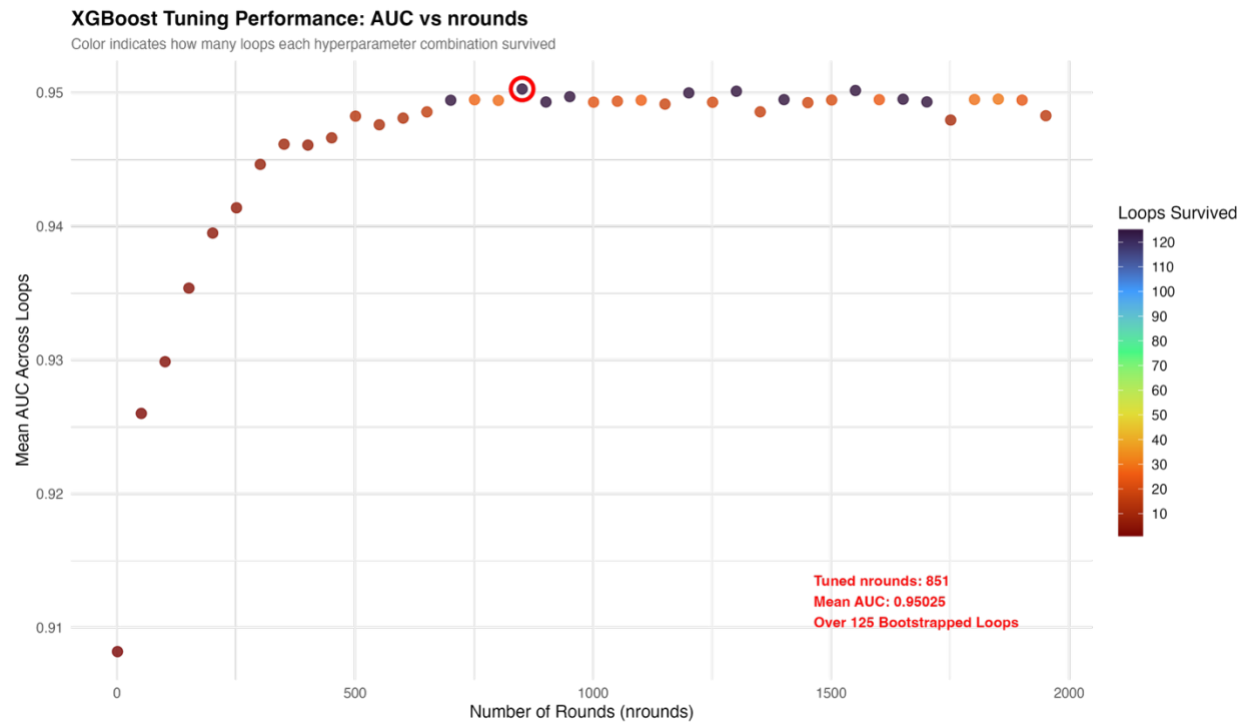


Figure 2

Our algorithm tunes nrounds to 851 and thus when we may run our final tests with nrounds set to 851 on 1000 bootstrapped samples where we receive a mean AUC of 0.949486 and a SD of 0.0027.

References

Burke, B. 2009. “Expected Point Values”. URL.

<https://www.advancedfootballanalytics.com/2009/12/expected-point-values.html>

Carl S, Baldwin B, Sharpe L, Ho T, Edwards J (2023). nflverse: Easily Install and Load the 'nflverse'. R package version 1.0.3, <<https://CRAN.R-project.org/package=nflverse>>.

Carter, V. and R. Machol. 1971. “Operations Research on Football”. *Operations Research* 19:541–544. [10.1287/opre.19.2.541](https://doi.org/10.1287/opre.19.2.541)

Chen, T., & Guestrin, C. (2016). *XGBoost: A scalable tree boosting system*. arXiv preprint arXiv:1603.02754. <https://doi.org/10.48550/arXiv.1603.02754>

Das, M., Newey, W. K., & Vella, F. (2003). Nonparametric Estimation of Sample Selection Models. *The Review of Economic Studies*, 70(1), 33–58.

<http://www.jstor.org/stable/3648610>

ESPN Staff. (2025, August 13). Highest-paid NFL players: Most guaranteed money at every position. *ESPN*. https://www.espn.com/nfl/story/_/id/34096853/highest-paid-nfl-players-tracking-most-money-guaranteed-per-year-every-position

Goff, B., & Locke, S. L. (2018). Revisiting Romer: Digging Deeper Into Influences on NFL Managerial Decisions. *Journal of Sports Economics*, 20(5), 671-689.

Heckman, J. J. (1979). Sample Selection Bias as a Specification Error. *Econometrica*, 47(1), 153-161. <https://doi.org/10.2307/1912352>

Newey, W. K. (2009). Two-step series estimation of sample selection models. *The Econometrics Journal*, 12(S1), S217–S229. <http://www.jstor.org/stable/23116602>

- Owens, M. F., & Roach, M. (2018). Decision-making on the hot seat and the short list: Evidence from college football fourth down decisions. *Journal of Economic Behavior & Organization*, 148, 226-245. <https://doi.org/10.1016/j.jebo.2018.02.023>
- Pro Football Focus (PFF). (2024). NFL Player Performance Data. Accessed via subscription at <https://premium.pff.com/nfl/teams/2024/REGPO>
- Pro-Football-Reference. (2024). 2024 NFL attendance data. *Pro-Football-Reference.com*. <https://www.pro-football-reference.com/years/2024/attendance.htm>
- Pro Football Focus. (n.d.). *Learn about PFF player grades*. <https://www.pff.com/grades>
- Romer, D. (2006). Do Firms Maximize? Evidence from Professional Football. *Journal of Political Economy*, 114(2), 340-365. <https://doi.org/10.1086/501171>
<https://doi.org/10.1177/1527002518798686>
- Yam, Derrick, and Michael Lopez. "Quantifying the causal effects of conservative fourth down decision making in the national football league." *Available at SSRN 3114242* (2018).
- Yurko, R., Ventura, S. & Horowitz, M. (2019). nflWAR: a reproducible method for offensive player evaluation in football. *Journal of Quantitative Analysis in Sports*, 15(3), 163-183. <https://doi.org/10.1515/jqas-2018-0010>